

Robin Liu

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Education

- Ph.D. University of California, Santa Barbara**, Statistics Santa Barbara, CA
2026
- Topics in joint estimation of the mean and covariance structure in high-dimensional data
 - Advisor: Prof. Guo Yu
- BS University of Michigan**, Honors Mathematics, Computer Science Ann Arbor, MI
2013

Experience

- Uniswap Labs**, Research Scientist New York City
Aug 2026 – present
1 month
- Uniswap Labs**, Ph.D. Research Fellow New York City
Aug 2025 – Feb 2026
7 months
- Developed and deployed the competitiveness index to compare DeFi protocols
 - Drew insights from large databases of transactions and A/B test results
 - Created shareable software and data resources used by data scientists and researchers
- Inria Centre**, Research Engineer University Grenoble Alpes
Summer 2024
- Developed scalable tools for fitting statistical mixed-effects models to brain imaging data
 - Leveraged distributed high-performance computing clusters to analyze large datasets
 - Accelerated computations by 300% through detailed analysis and simplification of numerical operations
- Quantitative Risk Management Inc.**, Software Developer Chicago
2013 – 2020
7 years
- Developed the software framework for assessing interest rate, liquidity, and other risks associated with asset and liability management
 - Designed and maintained database systems for complex financial products
 - Built a system for cleaning and aggregating financial transaction data using K-means clustering
 - Enhanced the portfolio optimization engine to support haircut modeling
- Spot Trading LLC**, Software Development Intern Chicago
Summer 2012
- Developed the trade management system of a proprietary options trading firm
 - Implemented low-latency trade execution on the Boston Options Exchange

Projects

- Deep residual networks for crystallography** SLAC National Accelerator Laboratory
- Developed a deep learning model in PyTorch for X-ray crystallography experiments
 - Trained CNNs for regression and classification of experimental results
 - Implemented transfer learning with ImageNet to improve prediction accuracy by 10%

Publications

- Estimation of the error structure in multivariate response linear regression models** 2025
- Liu, R.**, Yu, G.
[10.1002/wics.70021](https://doi.org/10.1002/wics.70021) (WIREs Comput Stat, 17: e70021)

Deep residual networks for crystallography trained on synthetic data Mendez, D., Holton, J.M., Lyubimov, A.Y., Hollatz, S., Mathews, I.I., Cichosz, A., Martirosyan, V., Zeng, T., Stofer, R., Liu, R. 10.1107/S2059798323010586 (Acta Crystallographica Section D: Structural Biology)	2024
Convex estimation of Gaussian graphical regression models with covariates Liu, R. , Yu, G., <i>Submitted</i>	2026
A mixed model approach for estimating regional functional connectivity from voxel-level BOLD signals Liu, R. , Zhang, C., Tran, C., Achard, S., Meiring, W., Petersen, A., <i>Submitted</i>	2026

Selected Honors

- 2024 WNAR Most Outstanding Paper Award
- 2024 UCSB PSTAT Departmental Travel Grant
- 2024-25 UCSB Doctoral Student Travel Grant

Skills

Languages: Python, R, Julia, C++, C#, SQL

Packages: PyTorch, Jupyter, SciPy, tidyverse

Technologies: .NET, Linux, Docker, BigQuery, Google Cloud Platform, Apache Spark, Databricks

Statistical and machine learning: Proficiency with deep learning, xgboost, and other nonlinear prediction methods. Proficiency with classical methods such as OLS, LASSO, multivariate statistics, covariance estimation, time series analysis.

Service

- Reviewer for *Nature Scientific Reports*

Presentations

A mixed model of regional functional connectivity from voxel-level BOLD signals

- WNAR/IMS Annual Meeting 2025; Whistler, BC

Covariate-adjusted Gaussian graphical models via natural parametrization

- WNAR/IMS/Graybill Annual Meeting 2024; Fort Collins, CO
- Joint Statistical Meetings 2024; Portland, OR
- CFE-CMStatistics 2024; London, UK
- Inria statistical research seminar 2024; Grenoble, FR

Teaching

Lead instructor: (PSTAT 10) Principles of Data Science

Teaching assistant: (PSTAT 232) Computational Statistics, (PSTAT 231) Statistical Machine Learning, (PSTAT 234) Statistical Data Science, (PSTAT 235) Big Data Analytics, (PSTAT 120B) Probability and Statistics II